

ROS Notes

Commands

General

- `rospack list`
 - Lists installed ROS packages
- `rospack find <package-name>`
- `rosls <package-name>`
 - ROS (directory) + ls
- `roscd <package-name>`
 - ROS (directory) + cd
- `roscore`
 - Starts the master node
- `roslaunch <package-name> <executable-name>`
 - Starts a node
 - Both params are required
 - `__name:=<node-name>`
 - Changes name of a node
- `rqt_graph`
 - Visualize publish-subscribe relationships between nodes
 - r = ROS, qt = Qt GUI,
- `roswtf`
 - Performs a variety of sanity checks

Node

- `roslaunch list`
 - Lists running nodes
- `roslaunch info <node-name>`
 - Info on a running node
- `roslaunch kill <node-name>`
 - Kills a running node
 - Better than ctrl + c since that might not remove it from the running node list.
 - `roslaunch cleanup` to fix

Topic

- `rostopic list`
 - Lists active topics
- `rostopic echo <topic-name>`
 - Topic echo
- `rostopic info <topic-name>`
 - Info on a active topic
- `rostopic hz <topic-name>`
 - Topic frequency
- `rostopic bw <topic-name>`
 - Topic bandwidth

Package

- `catkin_create_pkg <package-name>`
 - Create new ROS package
 - Creates a directory containing `package.xml` and `CMakeLists.txt`
- `add_executable(<executable-name> <source-files>)`

Dependencies

- `find_package(catkin REQUIRED COMPONENTS <package-names>)`
 - Add dependency to `CMakeLists.txt`
- `add_executable(<executable-name> <source-file>)`
- `target_link_libraries(<executable-name> ${catkin_LIBRARIES})`
 - Create an executable from source(s)
- `<build_depend>package-name</build_depend>`
- `<run_depend>package-name</run_depend>`
 - Add dependency to `package.xml`
 - Necessary for distribution

Notes

Book

- <https://www.cse.sc.edu/~jokane/agitr/>
- Nodes communicate via messages
 - Messages are categorized and organized by topics
 - Nodes will publish messages to a topic to “send”
 - Other nodes will subscribe to said topic to “receive” messages
- Message types are formatted as follows
 - `package-name/type-name`

- ROS nodes should be loosely coupled.
 - Shouldn't know about any others.
 - Use services for 1-1 communications
- ROS packages should only contain lowercase letters, digits, and underscores
 - First character has to be a letter
- Catkin is the ROS build system
- `ros::Rate` is much better than `sleep` or something since it accounts for the time spent performing computation in the loop.
 - Perfect for making sure inference runs at a specified fps

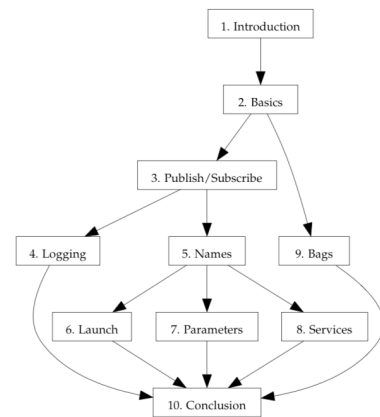


Figure 1.1: Dependencies between chapters.